

Can the disclosed number be reproduced?

A reviewer workbench that computes explicit evidence and escalates the rest.

The buyer problem: energy and emissions figures sit inside filings with different layouts, units, factor vintages, and levels of detail. Reviewers lose time determining which arithmetic can be reproduced and which claims need follow-up.

The useful result: this working prototype preserves a narrow source locator and hashes, converts supported units with decimal arithmetic, checks one pinned factor set, and returns **RECONCILED**, **MISMATCH**, or **REVIEW_REQUIRED** with a reason.

2
RECONCILED

1
MISMATCH

3
REVIEW REQUIRED

6/6
ORACLE CASES

Actual working-output screenshot

CO2

CARBON DISCLOSURE RECONCILIATION DESK

Runnable evidence / WS-004

REAL PYTHON ENGINE / BOUNDED FIXTURE

DISCLOSURE EVIDENCE REVIEW

Reproduce the math.
Escalate the gaps.

Compare declared energy and emissions arithmetic against an exact 2025 factor reference. The workbench computes only when the evidence supports it and routes ambiguity to human review.

01 / REVIEW INPUT

Run a reconciliation

Start with all six cases or operate the controlled unit-change scenario.

Evidence case

All six benchmark cases

Activity unit

MWh - baseline

Run evidence check

Restore six-case benchmark

Try the 1/1,000 unit change

PINNED FACTOR SET

UK GHG 2025 / V1 FINAL

Workbook updated 2025-06-10

Three minimized recorded filing sources plus two controlled scenarios. No company names or

02 / DECISION SUMMARY

Six evidence cases evaluated

Run ws004-1984fbedb2ae05ca / Python engine 2c7b693067...c125da / 6 cases

2
RECONCILED

1
MISMATCH

3
REVIEW REQUIRED

ORACLE REPRODUCED

03 / TRACEABLE OUTCOMES

Calculation and review queue

DECISION + REASON + SOURCE HASH

RECONCILED

Case 1 Filing Electricity Reconciled

Recorded Public Filing Fact

ACTIVITY

3,939,780 kWh

DISCLOSED

697.34 tCO2e

RECOMPUTED

697.34106 tCO2e

REASON

WITHIN TOLERANCE

CH-09904577-2025

SECR table / Purchased Electricity (Location based) / 2025

af67a7e729...e8ecfd

RECONCILED

Case 2 Controlled Mwh Reconciled

Reviewer Supplied Controlled Scenario

ACTIVITY

1,000 MWh

DISCLOSED

177 tCO2e

RECOMPUTED

177 tCO2e

REASON

WITHIN TOLERANCE

Evidence boundary: three minimized recorded filing sources plus two labeled controlled scenarios. No company names or personal fields are displayed. Results assess arithmetic reproducibility only.

The right factor year mattered more than the newest file.

Source inspection corrected the obvious approach before implementation.

The observed obstacle: preliminary feasibility selected the newer 2026 UK factor workbook. The three chosen filings explicitly cited 2025 factors. Comparing their arithmetic with 2026 values would create a confident-looking but false mismatch.

The implemented correction: freeze the official 2025 Version 1 / Final workbook, retain only the exact rows needed, hash both the source workbook and minimized factor subset, and refuse calculations when evidence does not identify a compatible basis.

BEFORE

Newest 2026 factor file selected during preliminary feasibility.

SOURCE CHECK

All selected filings reference the 2025 reporting factors.

AFTER

2025 V1 Final pinned by row ID, date, and SHA-256.

Actual code excerpt: compute only after evidence gates

```
if source.get("sourceFormat") not in SUPPORTED_SOURCE_FORMATS:
    return _review(result, "UNSUPPORTED_SOURCE_FORMAT")

if calculation.get("activityBreakdownAvailable") is False:
    return _review(result, "INSUFFICIENT_CALCULATION_EVIDENCE")

if (
    calculation.get("activityCategory") == "natural_gas"
    and not calculation.get("activityBasis")
):
    return _review(result, "AMBIGUOUS_ACTIVITY_BASIS")

required = (
    "activityValue",
    "activityUnit",
```

2025

MATCHED FACTOR YEAR

3

PINNED FACTOR ROWS

23/23

AUTOMATED TESTS

FAIL

CLOSED ON AMBIGUITY

Example: a natural-gas kWh value that does not say gross or net calorific value returns REVIEW_REQUIRED. The engine never chooses the factor that happens to match the disclosed result.

Change one unit. Watch the evidence decision change.

The controlled scenario makes the calculation and hash behavior independently testable.

1,000

BASELINE MWh

177.000BASELINE tCO₂e**0.177**CHANGED tCO₂e**23/23**

TESTS PASSING

Reviewer-operated proof: keep the numeric activity at 1,000 and change only the unit from MWh to kWh. Normalized activity and recomputed emissions become exactly 1/1,000 of baseline. The input hash changes; source, factor, and engine hashes remain stable; RECONCILED becomes MISMATCH.

BASELINE / MWh1,000,000 kWh normalized -> 177.000 tCO₂e -> RECONCILED**REVIEW CHANGE / kWh**1,000 kWh normalized -> 0.177 tCO₂e -> MISMATCH**Try it locally in about one minute:**

1. Run python app.py in the WS-004 project folder.
2. Open the local workbench link below; the six-case run executes automatically.
3. Click **Try the 1/1,000 unit change**.
4. Inspect reason codes and download the executed JSON and CSV.

Local working demo: 127.0.0.1:5000**Repository and reproducible evidence****Limits that matter**

Six bounded cases are not production accuracy. Recorded facts are minimized; controlled scenarios are labeled. No full filing bodies, company names, or personal fields are bundled. A result is not an audit opinion, compliance certification, environmental score, or statement that reported emissions are true.

Evidence run ws004-1984fbedb2ae05ca / changed run ws004-52cd149712424e77 / UK-GHG-2025-V1-FINAL